

Conectarse a AdventureWorksLT

```
SELECT TOP (5) CustomerID, FirstName, LastName
FROM SalesLT.Customer;
SELECT TOP (5) SalesOrderID, OrderDate, CustomerID
FROM SalesLT.SalesOrderHeader;
SELECT TOP (5) ProductID, Name, ListPrice
FROM SalesLT.Product;
```

Resultados		Mensajes	
CustomerID	FirstName	LastName	
1	Orlando	Gee	
2	Keith	Harris	
3	Donna	Cameras	
4	Janet	Gates	
5	Lucy	Harrington	
SalesOrderID	OrderDate	CustomerID	
1	2008-06-01 00:00:00.000	29847	
2	2008-06-01 00:00:00.000	30072	
3	2008-06-01 00:00:00.000	30113	
4	2008-06-01 00:00:00.000	29485	
5	2008-06-01 00:00:00.000	29957	
ProductID	Name	ListPrice	
1	HL Road Frame - Black, 58	1431.50	
2	HL Road Frame - Red, 58	1431.50	
3	Sport-100 Helmet, Red	34.99	
4	Sport-100 Helmet, Black	34.99	
5	Mountain Bike Socks, M	9.50	

Create view

```
CREATE OR ALTER VIEW SalesLT.vCustomerOrders AS
SELECT
    c.CustomerID,
    CONCAT(c.FirstName, ' ', c.LastName) AS CustomerName,
    h.SalesOrderID,
    h.OrderDate
FROM SalesLT.Customer c
INNER JOIN SalesLT.SalesOrderHeader h ON c.CustomerID = h.CustomerID;

SELECT TOP (5) *
FROM SalesLT.vCustomerOrders
ORDER BY OrderDate DESC;
```

Resultados		Mensajes		
	CustomerID	CustomerName	SalesOrderID	OrderDate
1	29847	David Hodgson	71774	2008-06-01 00:00:00.000
2	30072	Andrea Thomsen	71776	2008-06-01 00:00:00.000
3	30113	Raja Venugopal	71780	2008-06-01 00:00:00.000
4	29485	Catherine Abel	71782	2008-06-01 00:00:00.000
5	29957	Kevin Liu	71783	2008-06-01 00:00:00.000

Create a stored procedure to process an order

```
CREATE OR ALTER PROCEDURE dbo.AddOrderLineItem
    @SalesOrderID INT,
    @ProductID INT,
    @Quantity INT
AS
BEGIN
    SET NOCOUNT ON;
    BEGIN TRANSACTION;
    DECLARE @UnitPrice DECIMAL(18,2);
    SELECT @UnitPrice = CAST(ListPrice AS DECIMAL(18,2))
    FROM SalesLT.Product
    WHERE ProductID = @ProductID;
    IF @UnitPrice IS NULL
    BEGIN
        ROLLBACK TRANSACTION;
        THROW 50010, 'Invalid ProductID specified.', 1;
    END
    IF NOT EXISTS (SELECT 1 FROM SalesLT.SalesOrderHeader WHERE SalesOrderID =
@SalesOrderID)
    BEGIN
        ROLLBACK TRANSACTION;
        THROW 50011, 'Invalid SalesOrderID specified.', 1;
    END
    INSERT INTO SalesLT.SalesOrderDetail (SalesOrderID, OrderQty, ProductID, UnitPrice,
UnitPriceDiscount)
    VALUES (@SalesOrderID, @Quantity, @ProductID, @UnitPrice, 0);
    UPDATE h
    SET SubTotal = d.SumLineTotal,
        ModifiedDate = SYSUTCDATETIME()
    FROM SalesLT.SalesOrderHeader h
    INNER JOIN (
        SELECT SalesOrderID, SUM(LineTotal) AS SumLineTotal
        FROM SalesLT.SalesOrderDetail
        WHERE SalesOrderID = @SalesOrderID
        GROUP BY SalesOrderID
    ) d ON d.SalesOrderID = h.SalesOrderID;

    COMMIT TRANSACTION;
END;
DECLARE @SalesOrderID INT = (SELECT TOP 1 SalesOrderID
    FROM SalesLT.SalesOrderHeader
    ORDER BY SalesOrderID DESC);
EXEC dbo.AddOrderLineItem @SalesOrderID = @SalesOrderID,
    @ProductID = 680,
    @Quantity = 1;

SELECT TOP (5) *
FROM SalesLT.SalesOrderDetail
WHERE SalesOrderID = @SalesOrderID
ORDER BY SalesOrderDetailID DESC;
SELECT SalesOrderID, SubTotal, TaxAmt, Freight, TotalDue
FROM SalesLT.SalesOrderHeader
WHERE SalesOrderID = @SalesOrderID;
```

Resultados

Mensajes

	SalesOrderID	SalesOrderDetailID	OrderQty	ProductID	UnitPrice	UnitPriceDiscount	LineTotal	rowguid	ModifiedDate
1	71946	113407	1	680	1431.50	0,00	1431.500000	36BBD4EA-3BF6-4737-AAFE-781194013487	2026-09-21 11:50:57.513
2	71946	113406	1	916	31.584	0,00	31.584000	B571CEE0-A90A-4E76-9796-379B55352A28	2008-06-01 00:00:00.000

	SalesOrderID	SubTotal	TaxAmt	Freight	TotalDue
1	71946	1463,084	3,1163	0,9738	1467,1741

Crear funcion escalar

```
CREATE OR ALTER FUNCTION dbo.fnOrderTotal (@OrderID INT)
    RETURNS DECIMAL(18,2)
AS
BEGIN
    DECLARE @Total DECIMAL(18,2);

    SELECT @Total = SUM(LineTotal)
    FROM SalesLT.SalesOrderDetail
    WHERE SalesOrderID = @OrderID;
    RETURN ISNULL(@Total, 0.00);
END;
```

	SalesOrderID	OrderTotal
1	71946	1463.08
2	71938	74160.23
3	71936	79589.62
4	71935	5533.87
5	71923	96.11
6	71920	2527.13

Crear una inline table-valued function (TVF)

```
CREATE OR ALTER FUNCTION dbo.GetCustomerOrders (@CustomerID INT)
    RETURNS TABLE
AS
RETURN
(
    SELECT
        h.SalesOrderID,
        h.OrderDate
    FROM SalesLT.SalesOrderHeader h
    WHERE h.CustomerID = @CustomerID;
)
SELECT *
FROM dbo.GetCustomerOrders(29929)
ORDER BY OrderDate DESC;
SELECT CONCAT(c.FirstName, ' ', c.LastName) AS CustomerName, o.SalesOrderID, o.OrderDate
FROM SalesLT.Customer c
    CROSS APPLY dbo.GetCustomerOrders(c.CustomerID) o
WHERE c.CustomerID = 29929;
```

	SalesOrderID	OrderDate
1	71902	2008-06-01 00:00:00.000

	CustomerName	SalesOrderID	OrderDate
1	Jeffrey Kurtz	71902	2008-06-01 00:00:00.000

Trigger to log

```
IF OBJECT_ID('dbo.OrderAudit') IS NULL
BEGIN
    CREATE TABLE dbo.OrderAudit (
        AuditID INT IDENTITY(1,1) PRIMARY KEY,
        OrderID INT NOT NULL,
        OldTotal DECIMAL(18,2) NULL,
        NewTotal DECIMAL(18,2) NULL,
        ChangedAt DATETIME2 NOT NULL DEFAULT SYSUTCDATETIME()
    );
END
GO
CREATE OR ALTER TRIGGER SalesLT.trg_LogOrderTotalChange
ON SalesLT.SalesOrderDetail
AFTER INSERT, UPDATE
AS
BEGIN
    SET NOCOUNT ON;

    ;WITH AffectedOrders AS (
        SELECT SalesOrderID FROM inserted
        UNION
        SELECT SalesOrderID FROM deleted
    ),
    NewTotals AS (
        SELECT d.SalesOrderID, SUM(d.OrderQty * d.UnitPrice) AS Total
        FROM SalesLT.SalesOrderDetail d
        INNER JOIN AffectedOrders a ON d.SalesOrderID = a.SalesOrderID
        GROUP BY d.SalesOrderID
    ),
    InsertedTotals AS (
        SELECT SalesOrderID, SUM(OrderQty * UnitPrice) AS Total
        FROM inserted
        GROUP BY SalesOrderID
    ),
    DeletedTotals AS (
        SELECT SalesOrderID, SUM(OrderQty * UnitPrice) AS Total
        FROM deleted
        GROUP BY SalesOrderID
    )
    INSERT INTO dbo.OrderAudit (OrderID, OldTotal, NewTotal)
    SELECT
        n.SalesOrderID,
        n.Total - ISNULL(i.Total, 0) + ISNULL(d.Total, 0) AS OldTotal,
        n.Total AS NewTotal
    FROM NewTotals n
    LEFT JOIN InsertedTotals i ON n.SalesOrderID = i.SalesOrderID
    LEFT JOIN DeletedTotals d ON n.SalesOrderID = d.SalesOrderID;
END;

UPDATE d
SET OrderQty = OrderQty + 1
FROM SalesLT.SalesOrderDetail d
WHERE d.SalesOrderID = (SELECT TOP 1 SalesOrderID FROM SalesLT.SalesOrderHeader ORDER BY SalesOrderID
DESC);

SELECT TOP (5) *
FROM dbo.OrderAudit
ORDER BY AuditID DESC;
```

Resultados		Mensajes			
	AuditID	OrderID	OldTotal	NewTotal	ChangedAt
1	1	71946	1463.08	2926.17	2026-09-21 10:33:21.5842048